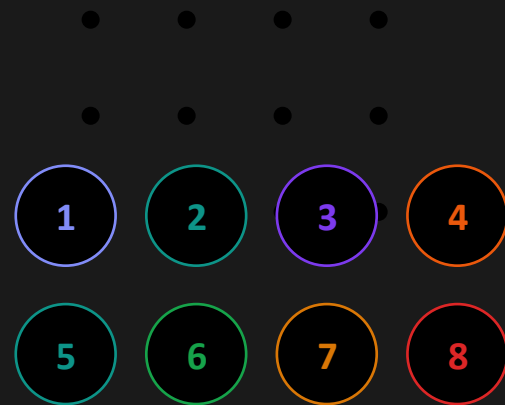


AI Architecture Advisor

Converts a 7-step discovery session into a complete enterprise AI architecture package — the same deliverables a Principal Architect takes a week to produce, in 60 seconds.



Enterprise AI architecture starts from scratch — every single time



Discovery insights get lost in handoffs

A senior architect runs a discovery session, produces notes and slide fragments, then hands off to a different team member to build the architecture. Context evaporates. The next engagement starts from scratch.

Result: inconsistent architecture quality across engagements



ADRs never get written until an audit forces it

Architecture Decision Records are the formal artifact that explains why a system was built a specific way. They are almost never written during the design phase — only retroactively, under pressure, when something goes wrong.

Result: technical debt with no documented rationale



The value case is built after the architecture decision

ROI models and CFO narratives are assembled after the architecture is already chosen — making them advocacy documents, not analysis. By then the team is defending a decision, not evaluating one.

Result: investment approvals slow down or fail entirely

7 enterprise deliverables · 60 seconds

01 Reference Architecture

Layered target-state diagram with specific technologies

Replaces: Architecture design

02 Architecture Decision Records

Context · decision · rationale · tradeoffs · alternatives

Replaces: ADR authoring

03 Risk Register & Controls

Risks ranked by severity with governance controls

Replaces: Risk assessment

04 30/60/90 Day Roadmap

Phased milestones, dependencies, team structure

Replaces: Program planning

05 TCO / ROI / NPV Model

Full financial model with sensitivity at 65/100/125%

Replaces: Value engineering

06 CFO Executive Narrative

Decision-ready one-pager with investment justification

Replaces: Business case writing

07 AI Readiness Scorecard

6-dimension score with gap analysis and legend

Replaces: Maturity assessment

Most demos produce chat responses.

This produces Architecture Decision Records.

What is an ADR?

- ◆ An Architecture Decision Record documents a significant design choice — the context that drove it, the decision made, the rationale, the tradeoffs accepted, and the alternatives rejected.
- ◆ ADRs outlast the engineers who wrote them. They answer the question 'why was this built this way?' years later — during audits, migrations, and onboarding.
- ◆ Producing ADRs requires deep technical judgment and domain context. Most AI demos don't come close.
- ◆ This tool generates 4-5 production-quality ADRs per blueprint, grounded in the discovery inputs and architecture decisions.

ADR-002 LLM Provider Selection Proposed

Context	Regulated telco environment requires data residency in Azure. Single LLM provider creates availability risk for critical NOC operations requiring 24/7 AI assistance.
Decision	Primary: Azure OpenAI gpt-4o. Fallback: Anthropic Claude claude-haiku-4-5 via Azure Marketplace.
Rationale	Azure OpenAI meets data residency compliance. Dual-provider fallback ensures 99.9% AI availability with seamless failover using identical prompt format.
Pros	Data sovereignty compliance · High availability via fallback · Azure native billing and audit
Tradeoffs	Prompt compatibility testing required across both providers · Cost monitoring across two bills
Rejected	OpenAI direct API: data leaves Azure boundary · Single provider: unacceptable availability risk

Three clean layers · browser-native · no server required

PRESENTATION LAYER

Browser-native · GitHub Pages deployable · zero infrastructure cost

- ◆ index.html — mode selector, scenario picker, wizard shell, results tabs
- ◆ assets/css/styles.css — full design system, dark theme
- ◆ Full-screen wizard overlay · Agent pipeline animation · 7-tab results

Zero infrastructure
GitHub Pages in 60s

ORCHESTRATION LAYER

JavaScript · sequential agent loop · no external framework

- ◆ app.js — mode, pipeline, demo runner, export
- ◆ AGENT_REGISTRY[] — ordered array, 10 lines
- ◆ Prior results accumulated and passed to each agent
- ◆ FinancialModel.calculate() runs before Agent 08

Deterministic sequential
Browser JavaScript

AGENT + API LAYER

8 independent agent files · dual-provider client · JSON repair

- ◆ agent-01-discovery.js → agent-08-value.js
- ◆ api-client.js — Claude ↔ OpenAI, 1-line swap
- ◆ demo-blueprint.js — pre-built output, no API needed
- ◆ scoring.js + financial.js — pure math, no API dependency

Claude or OpenAI
1-line swap

Agent anatomy · prior results chaining · provider abstraction

Each agent file — e.g. agent-04-architecture.js

`buildPrompt(answers, prior)`

Assembles the prompt from wizard answers + all prior agent outputs. Each agent sees the full accumulated context chain.

`async run(apiKey, answers, prior)`

Calls `buildPrompt()` → `ApiClient.call()` → `ApiClient.parseJSON()`. Returns a typed schema object specific to that agent.

Output schema

Each agent returns a defined JSON structure. Agent 04 returns architecture layers + ADRs. Agent 06 returns risk register + controls.

api-client.js — Provider abstraction

`window.ApiClient.init('claude')` or `window.ApiClient.init('openai')`

One line in `app.js`. All 8 agent files unchanged. Provider handles endpoint, headers, body format, and response parsing automatically.

Prior results chaining — agents build on each other's outputs

Agent 01 — Discovery

IN: Wizard answers only

OUT: `problemStatement` · `rootCauses` · `stakeholderMap`

Agent 04 — Architecture

IN: Answers + `prior[discovery]` + `prior[capability]`

OUT: `architectureLayers` · `adrs` · `integrationPatterns`

Agent 07 — Roadmap

IN: Answers + `prior[governance]` · `riskRegister` + `prior[data].gaps`

OUT: `roadmap` · `goLiveCriteria` · `teamStructure`

Agent 08 — Value

IN: Answers + `prior[ALL]` + pre-calculated financials

OUT: `cfoNarrative` · `investmentJustification` · `advisoryPath`

JSON repair chain — 5 attempts, handles token-limit truncation

1. Direct `JSON.parse()`
2. Strip non-printable control characters
3. Collapse literal newlines inside strings
4. Collapse whitespace runs
5. Close unclosed brackets — handles truncated responses

Mode → Scenario → 7-step wizard → 8 agents → 7 deliverables

1

10 seconds

Select mode & scenario

Demo (no key), Claude, or OpenAI.
Choose from 15 pre-built scenarios.
Wizard auto-fills with complete
discovery answers.

2

30–60 seconds

Review 7-step wizard

Adjust pre-filled answers or fill from
scratch. Profile · Problems · Outcomes
· Capabilities · Data · Governance ·
Value.

3

8–90 seconds

Watch 8 agents run

Each agent animates in sequence.
Demo Mode: 8 seconds. API Mode:
45–90 seconds of real LLM reasoning.

4

Immediately

7 enterprise deliverables

All 7 tabs populate simultaneously.
Export as Markdown or JSON. Share
with the team.

Zero setup to live AI generation — one click apart



No API key required

Demo Mode

Select Demo → pick scenario → click Generate. Pre-built Telco NOC blueprint loads in 8 seconds. All 8 agents animate. All 7 tabs populate with real content.

Best for: Always ready · no setup · works offline · no network dependency



Anthropic · claude-haiku-4-5

Claude API

Switch to Claude → enter sk-ant-... key → run. 8 real API calls. Results grounded in actual LLM reasoning for the specific discovery inputs provided.

Best for: Deepest output quality · most expressive · production use



OpenAI · gpt-4o-mini

OpenAI API

Switch to OpenAI → enter sk-... key → run. Identical 8-agent pipeline, same output structure. Provider swap is one line in api-client.js.

Best for: OpenAI ecosystem · cost-effective · A/B comparison

15 SCENARIOS

Built for every enterprise vertical

01

Telco Network AIOps

Telecom

02

Financial GenAI Assistant

FinServ

03

Healthcare Clinical AI

Healthcare

04

Government Doc Intelligence

Government

05

Enterprise ITSM (ServiceNow)

IT Ops

06

Retail Customer Experience

Retail

07

Manufacturing Pred. Maint.

MFG

08

Cybersecurity Threat Intel

SecOps

09

Insurance Claims AI

Insurance

10

Energy Grid Optimization

Energy

11

Legal Contract Intelligence

Legal

12

Supply Chain Intelligence

Logistics

13

HR Talent Intelligence

HR

14

Pharma Clinical Trials AI

Pharma

15

Private Cloud AI Infra

Platform

Real output · real numbers · Telco NOC scenario

\$845K**Annual gross benefit**

Labor + rework + delay savings

87%**3-year ROI**

Net benefit / total cost

\$1.1M**NPV (8% discount)**

3-year net present value

14 mo**Payback period**

Break-even on implementation

\$2.6M**Cost of inaction**

3-year status quo cost

Calculation — Telco NOC · 12,000 incidents/yr · \$95/hr loaded labor

Annual labor cost (current)	\$630K 12,000 × 35 min × \$95/hr
Annual rework / escalation	\$132K 8% error rate × \$300 avg
Annual delay cost	\$180K Escalation and SLA exposure
Labor savings (35% gain)	\$220K Productivity improvement
Rework savings (25% reduction)	\$33K AI-grounded recommendations
Implementation cost	\$250K One-time
Annual operating cost	\$90K Platform + support

Sensitivity analysis

Conservative (65%)

\$549K/yr · 42% ROI · 22 mo payback

Expected (100%)

\$845K/yr · 87% ROI · 14 mo payback

Optimistic (125%)

\$1.06M/yr · 132% ROI · 11 mo payback

LIVE DEMO AVAILABLE

Try it right now

No API key required

[Live demo → danvzla.github.io/ai-architecture-advisor](https://danvzla.github.io/ai-architecture-advisor)

[GitHub repo → github.com/danvzla/ai-architecture-advisor](https://github.com/danvzla/ai-architecture-advisor)

- ◆ Demo Mode → select any scenario → Generate Blueprint → 7 tabs in 8 seconds · no API key
- ◆ 15 scenarios · 8 agents · 7 enterprise deliverables · Claude API · OpenAI API
- ◆ Produces: Architecture · ADRs · Risk Register · Roadmap · ROI Model · CFO Narrative

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